The Timing of Teenage Births: Estimating the Effect on High School Graduation and Later Life Outcomes

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Abstract

We examine the long-term outcomes for a population of teenage mothers who give birth to their children around the end of their high school year. We compare the mothers whose high school education was interrupted by childbirth, because the child was born before her expected graduation date to mothers who did not experience the same disruption to their education. We find that mothers who give birth during the school year are seven percent less likely to graduate from high school, are less likely to be married, and have more children than their counterparts who gave birth just a few months later. The labor market outcomes for these two sets of teenage mothers are not statistically different, but with a lower likelihood of marriage and more children, the households of the treated mothers are more likely to fall below the poverty threshold. While differences in educational attainment have narrowed over time, the differences in labor market outcomes and family structure have remained stable.

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1 Introduction

There is an extensive body of literature establishing a strong correlation between teenage childbearing and poor economic outcomes later in life. Teenage mothers are less likely to complete high school, less likely to be working, have lower earnings and are less likely to be married than those women who did not become teenage mothers (Card and Wise, 1978; Trussell, 1988; Ellwood, 1989). However, there is also evidence that the young women who have children as teenagers come from less advantaged backgrounds than those who delay childbearing until later in life. This makes it very difficult to determine whether having a child as a teenager caused the poor economic outcomes or whether they are symptoms of the less advantaged upbringing that is also correlated with these outcomes. In fact, the few papers that have used more sophisticated identification methods than comparing teenage mothers with those who delayed childbearing, have found that the effect of teenage childbearing on economic outcomes is small (Geronimus and Korenman, 1992; Hotz et al., 2005; Ashcraft et al., 2013).[1]

[1]Geronimus and Korenman (1992) control for family background by comparing pairs of sisters, where one has a child as a teen and the other does not and find that studies that do not control for family background overstate the causal consequences of teenage childbearing. Both Hotz et al. (2005) and Ashcraft et al. (2013) study this issue by using miscarriages as a source of identification. Hotz et al. (2005) use miscarriages as an instrument and find very small, short lived negative effects for most examined outcomes. For annual hours of work and earnings, they find that the teenage mothers are actually doing better at older ages than they would have been if they had delayed childbearing. Ashcraft et al. (2013) show that IV underestimates the negative effect of childbearing, but that OLS overestimates it. Using both estimates, they derive a consistent estimator of the effect of giving birth for those teens who would choose not to abort and find small negative effects of teenage childbearing.

One clear difference between women who have their first child as teenagers, and women who do not, is that a teenage mother's high school experience is more likely to be interrupted by the arrival of their child. A common policy prescription for improving the economic wellbeing of teenage mothers and their families is to make sure that the mothers have the necessary support to graduate from high school, and we aim to shed light on whether this type of policy is likely to have the intended effect. We look at the effect of having a child during high school versus becoming a young mother, but one who has already finished high school. Specifically, we compare the outcomes of women who had a child near the end of their senior year of high school to those who have one just after the (expected) end of high school.[2]

We find that women whose high school education was interrupted are 7 percent less likely to graduate from high school, relative to the mean, than those who had their first child a few months later. We find that the interruption of schooling affects later life outcomes of the teenage mothers. Young women who have a child prior to their expected graduation date are slightly less likely to be working or married, and have a higher number of children, on average. There is no statistically significant difference in earnings or the probability of working for the women whose education was interrupted. As a result of having more children and a lower likelihood of being married, they and their families are more likely to fall below the poverty line.

Given that the sizable difference in educational attainment has no estimated effect on wage income, our result differs from the majority of papers that estimate the signaling value of a high school degree and find (sometimes large) positive effects (see for example Jaeger

[2]In classifying women into groups, we assume normal school progress. "Just after the end of high school" means just after they would have graduated, whether they did or not. This will be discussed in more detail in Section 2.

and Page, 1996; Tyler et al., 2000), but it is consistent with Clark and Martorell (2014), who find little evidence of a signaling effect when comparing high school seniors who score just above the passing threshold on their high school exit exam to those who score just below. Knowing the difference in the value of a degree for this population is important for policy makers who have worked hard to increase the graduation rate for teenage mothers. For example, one of the benefits of Title IX of the 1972 Educational Amendments to the Civil Rights Act (Title IX), which prohibited discrimination on the basis of gender, was that it increased the high school graduation rates for teenage mothers (Guldi, 2016). Knowing that this, alone, is not sufficient to improve labor market incomes is important.

2 Data

This paper takes advantage of the fact that the timing of birth within the teenage years might matter for future outcomes. In particular, we look at a tight band of time around high school graduation. The main comparison examines the differences in outcomes for women who had their first child in January through June of their senior year of high school, and women who had their first child shortly after the end of their senior year: between July and December.[3] The main identifying assumption is that by comparing two groups of teenage mothers, who differ in the timing of their births by only a few months, there are no differences between the two groups prior to childbearing. Section 4 will provide evidence that this is the case.

The main analysis uses the population of 20-35 year old mothers from the 1980 and 2000 Censuses as well as the American Community Survey (ACS). One benefit of using Census and ACS data is that this gives us a large sample size. Our main treatment and control groups include more than 450,000 teenage mothers. The Census data come from the long-

[3]In a robustness check, we also consider a smaller window, where the treatment group had their first child in April-June and the control group in July-September.

form of the Decennial Census, a 1 in 6 sample of U.S. households that were asked a longer series of questions that included education, labor force status, and income. The ACS is an ongoing survey of approximately 3 million households annually. The data we use come from the 2005-2014 ACS surveys. The sample is limited to women who are ages 20-35 at survey. In order to observe that a woman is a mother, her child must be living in the same household when surveyed. The youngest mothers in the treatment group were 17 years old when they gave birth, meaning that their children turn 18 when they are 35 years old. Figure 1 shows the relationship between the timing of the birth of the child and the date of the survey. We limit the sample to mothers who are at least 20 years old by the survey date to abstract from the end of high school, thus excluding those that became teenage mothers within 2-3 years of the survey date.

We use information on the relationship of the household members to the household head to identify mothers and children in each survey and use the survey date to back out the age of the mother when she gave birth. The 1980 Census includes age (on the survey date: April 1, 1980) and quarter of birth for both the mother and child.[4] Using this information, and

[4]Because year of birth is not available, survey date is important for calculating each mother's age when she gave birth. For example, imagine a mother who was born on April 15th, 1950 (Q2) and has a child who was born on May 1, 1965 (Q3). On survey day, the mother is 29, and her child is 14 (neither one has had their birthday yet). Here, it is easy to see that the mother was 15 years old when her child was born. Now, imagine that the mother was born on March 1st, 1950 (Q1). In this example, the mother has already turned 30 by April 1st, but the child is still 14. Knowing that April 1 falls between their birthdays allows us to accurately calculate that the mother was 15 years old when her child was born, not 16.

the age of a mother's oldest "own child" living with her at survey date,[5] we calculate each mother's age when she gave birth to her first child. The 2000 census dataset and the ACS includes birthdate for both the mother and child, which allows us to calculate the mother's age when the child was born without making assumptions regarding when in the quarter mother and child were born, but we use consistent treatment and control groups to the 1980 matched sample, breaking at the quarter, to allow for comparability of results.

Figure 2 shows how women are assigned to groups. To assign women to the treatment and control groups, we assume that the cutoff for starting school is October 1st (the first day of the third quarter of the calendar year) and that everyone makes normal school progress.[6] In the 1980 census, where we do not know the exact birthdate, we also assume that if the mother and child are born in the same quarter, the mother's birthday is first. The shaded rows in the first two columns represent the treated group, while the shaded rows in the 3rd and 4th columns represent the control group. Each cell displays the school year at ages 17, 18 and 19 for someone who has made normal progress in school. For example, if a woman was born in Q1, had her first child in Q1, and was 18 years old when she had the child, she became a mother during January-March of her senior year of high school. She is assigned to the treatment group. However, if a woman was born in Q1, had her first child in Q3, and was 18 when she had the child, she became a mother during the summer after her senior year of high school and is assigned to the control group.

[5]We only include children who are 18 or younger at survey date.

[6]We do this out of necessity (retrospective information on school progress at the time of giving birth is not available in the data), but also to abstract from any endogenous differences in the school progress. By imposing normal school progress on all mothers, we introduce some classical measurement error, but do not introduce any additional bias stemming from endogenous differences stemming from the pace of school progress.

Table 1 displays the summary statistics for the treatment and control groups, as well as for a sample of women who delayed childbearing until they were between 23-25 years old.[7] Research has found evidence of seasonality in the types of women giving birth over the year (Buckles and Hungerman, 2013), so these women will be used to difference out any seasonality. Women who had their first child between Jan-June are "Treated" and women who had their first child between July-December are the "Control" group. Among the teen mothers, we can see that the treatment group is slightly younger at survey date, and a higher percentage of the group is black. This highlights the importance of including controls for age and race in the main specification.

The table also gives a first look at the main outcome variables of interest. The stars in the first column indicate a statistical difference between means for the treatment and control groups in the teenage cohort.[8] The treated group has fewer years of completed education and is less likely to have completed high school.[9] They are also less likely to be married, have more children, have slightly lower family income levels, and are more likely to fall below the poverty line.[10]

Figure 4 gives another initial picture of our main outcome variables of interest. The figure shows the average value of the education variables for women that had their first children in the months surrounding their expected graduation date. While these figures do

[7]We pick 23-25 because most women will have completed their education by this age. In a robustness check, we also consider a younger control group.

[8]These are simple differences without controls or clustered standard errors.

[9]"HS Degree" is defined as having completed at least 12 years of education, or having obtained a General Educational Development Test (GED).

[10]Summary statistics that have been compiled separately for each survey are available in Tables A1 - A3.

not include any of the control variables that will be described in Section 3, and only include women surveyed in the 2000 Census and the ACS, they help us understand the nature of the "first stage" variation (the effect of timing of birth on education). While we see that the probability of obtaining a high school degree increases leading up to and following expected high school graduation, we see do not see a break in trend or discontinuity. This suggests that the "treatment" is more continuous than one might expect. The timing of births is important for the probability of completing high school, but we do not see evidence that one particular time during this period around high school graduation matters most.

While important for interpreting our results, this is perhaps unsurprising for a few reasons. First, all women observed in the 2000 census and ACS samples completed high school after Title IX, which required schools to provide equal access to education for girls, even if they were pregnant or had children (Guldi, 2016). So, it is very unlikely that schools would force girls to drop out or purposefully make attending more difficult; doing so would put their federal funding at risk. Second, while giving birth is likely to disrupt schooling, having an infant at home continues the potential disruption. So, the earlier the birth, the longer the mother must balance school and motherhood in order to graduate. Similarly, the initial disruption might occur when the mother discovers she is pregnant. Even for our control group, this realization occurs during the school year, and it might affect her likelihood of graduating. Finally, our measure of high school graduation includes those who received their GED. Even if a mother drops out of high school, her likelihood of successfully completing a GED is expected to increase with the length of time she remained in school.

However, there *is* a break in the probability of having some college education or a college degree. We can see that for those giving birth after expected graduation, the slopes become steeper. Furthermore, it is interesting to note that while they are part of our control group, those giving birth during the summer between high school and college are even less likely to begin or complete college than those who give birth at the end of their final year of high

school. This suggests that with respect to college attendance, this is a crucial time period.

Again, these graphs do not include any control variables, and the next section describes our estimation strategy that uses this timing to better identify the effect of the timing of teenage pregnancy on educational and other outcomes.

3 Estimation Strategy

In order to estimate the effect of an interruption in high school education on our outcomes of interest, we estimate the following equation, first for the full sample and then separately for white, black and hispanic mothers:

$$\begin{aligned} \text{Outcome}_{isarc} &= \alpha + \beta_1 \text{Treat}_{isarc} \times \text{Teen}_{isarc} + \beta_2 \text{Treat}_{isarc} + \beta_3 \text{Teen}_{isarc} \\ &\quad + \Phi_s + \Phi_a + \Phi_r + \Phi_c + \varepsilon_{isarc} \end{aligned} \tag{1}$$

where Treat_{isarc} is an indicator variable equal to one if individual i's child was born January through June, and Teen_{isarc} is an indicator variable equal to one if the mother had her first child as a teen. The coefficient of interest is β_1, which measures the difference in differences for the outcome variable. It gives the effect of giving birth just before the end of high school, rather than just after the end of high school, with seasonality differenced out by the sample of older mothers. Additionally, all regressions include a full set of fixed effects for state Φ_s, mother's age at survey Φ_a mother's race Φ_r and census year Φ_c. Standard errors are clustered by state.

We examine a number of outcome variables. First, we look at whether there is a difference in years of education or in the probability that the mother has completed at least 12 years of

school.[11] Next, we estimate whether the interruption in education has detrimental effects on income or on the probability that her family falls below 100% or 200% of the poverty line. Then, we look at whether there are any differences in family structure at survey date. Does she have more children, is the age gap between the first and second child larger or smaller, and is she more or less likely to be married? Finally, we investigate whether the differences we observe in educational attainment and family structure affect educational outcomes for the children of the teenage mothers. We examine whether there are any differences in age for grade for the children.

4 Validity of the Identification Assumption

To validate our identification assumptions, we show that prior to their pregnancy, the young women in the two groups of teenage mothers, who differ in the timing of their births by only a few months, are not different from each other. In addition, within this small window we argue that the timing of the birth is exogenous, since most teenage pregnancies are unplanned. However, it is possible that the young women in the control group were planning to have a child just after graduation. Then, we would be concerned that much of our variation is driven by wanting/not wanting to have a child. The following analysis attempts to rule out any pre-birth differences and any differences in planning.

4.1 Natality Data

We use Natality Data from the National Vital Statistics System of the National Center for Health Statistics from the years 1969-2012 to look at seasonality in the demographics of

[11] This also includes women who have obtained a General Educational Development Test (GED).

women giving birth over the year.[12] The files contain a mix of either a 50% or 100% sample of births (depending on state and year). The records contain data on the mother's age, race, and number of prenatal visits.

Figure 3 shows the pattern of birthrates over the year for 17, 18 and 19 year olds as well as a sample of older women: 23, 24 and 25 year olds. Column 1 of Figure 3 shows that the pattern of births is largely cyclical for both age groups (more in the summer, fewer in the winter). We do not see a larger spike during the summer months for the high school girls than for the older ones, which might lead us to believe that the teens were consciously timing their pregnancies to have their first child after graduation. Furthermore, it shows that the older cohort should do a good job of differencing out seasonality in the regressions. We show the graph in two ways. The first row shows the pattern in birth rates over the year on the same scale for both the teen moms and the older moms. It is clear that while the birthrates are significantly higher for the older mothers, the pattern is very similar over the year. Row 2 shows the same graphs, but the teenage cohort is shown on the left axis, and the older cohort is on the right axis. This gives a better illustration of patterns over the year. Still, it is apparent that the seasonality in birthrates is very similar for both age groups.

In addition to overall birthrate, Figure 3 shows changes in the racial composition and average number of pre-natal visits over the year. Although there is seasonality in the racial composition of mothers, the pattern is very similar for the two cohorts. While the older cohort has more prenatal visits, on average, when compared to teenage cohort, there does

[12]These years were chosen to match the women surveyed in the 1980 census, where we find the largest differences in outcomes.

not appear to be strong seasonality for this measure for either group.[13]

4.2 National Survey of Family Growth

We use data from the National Survey of Family Growth to examine whether the treatment and control groups are similar on a number of additional measures, many of them determined pre-pregnancy. While this dataset is not ideal for the main analysis because of its small size and imprecise income data, it offers a number of interesting survey questions. The survey includes questions regarding sexual activity, sexual education and family background. The variables we use are defined in Table 2. For each variable, we estimate the difference-in-differences, using a sample of 23-25 year olds to difference out seasonality, as we do in the main results. As displayed in Table 3, the difference-in-differences is not statistically significant for any of the control variables tested. These results lend credibility to the assumption that these are comparable groups. That said, if one examines the signs of the coefficients, they point towards the treatment group being slightly less advantaged than the control group. If this is truly the case, we will be biased towards finding a negative effect of having an interruption during high school. It will bias both the education *and* the income coefficients, which are both expected to be negative, to be larger in magnitude.

[13]For all outcomes in Figure 3, we also ran a simple difference in differences model to test whether any of these outcomes were statistically different. We confirm that there are no statistical differences. These results, and more details, are available upon request.

5 Results

5.1 Education

The outcome variable that is most likely to be affected by the disruption in high school from a teenage pregnancy is education itself. This is the first-order effect. If there is no estimated effect on education, we would not expect to see an effect on other outcomes. Table 4 displays the results from estimating equation 1 for four measures of education. As expected, there are fairly large and statistically significant differences in both the probability of receiving a high school diploma, the probability of having some college education, and the total amount of completed education between the treatment and control groups. The results for a college degree are mixed.

Row 1 shows the effect on an indicator variable equal to one if the mother has a high school degree. The teens who have their child prior to their expected graduation date are 5.4 percentage points less likely to have finished high school than the control group, a 7 percent decrease relative to the mean. The effect is largest for white mothers and smallest for Hispanic mothers. As a percentage of the mean, treated white mothers are 8.5% less likely to complete high school, while black mothers are 5.2% less likely and Hispanic mothers are 5.9% less likely.

The second row shows the probability of completing some college education. Some college education is 4.7% lower for treated mothers, relative to the mean. Even though the timing of our variation is based on high school graduation date, the effect on college attendance is consistent with the expected effect of educational disruption. Most colleges require a high school diploma or GED for enrollment. Even if the teenage moms that miss their graduation date get a GED at a later date, they are less likely to have it in time for college enrollment than their counterparts that were able to finish the last several months of high school. The inconsistent results on college graduate imply that even those teenage moms who have their

children after their expected high school graduation date struggle to complete college. This is consistent with the pattern we observed in Figure 4. The precise timing that matters for college does not line up with expected high school graduation. Rather, those who give birth during the following summer are at least as affected as those who give birth during high school.

The last row displays the results for the mothers' total years of education. Column (1) shows that, on average, the treated teenagers completed 0.137 fewer years of education than the control group. Columns (2), (3) and (4) show the results stratified by race. The magnitude, when measured as a percent of the group specific mean, is largest for Hispanic mothers and smallest for black mothers for this measure of completed education.

5.2 Family and Labor Market Outcomes

We show that an interruption in high school due to childbirth decreases the probability of graduating from high school by 7 percent, the probability of attending college by 4.7 percent, and decreases total years of education by 0.137 years in Table 4. Tables 5 to 7 show the effects of this interruption on later life outcomes.

The biggest long term effect of high school disruption due to teen fertility is on family structure. Row 1 of Table 5 shows that the treated teenagers are less likely to be married, and row 2 shows that they have more children, on average, than the control group. These effects show up for all groups. White women also wait longer to have their second child than the control group, though this effect is smaller and not statistically significant for Black and Hispanic women.

Next, we look at whether these large differences in educational attainment translate into differences in the labor market. Row 1 of Table 6 shows that the treated mothers are less likely to be working, but the magnitude is small and the coefficients are not statistically significant for black or Hispanic mothers. Despite strong estimated effects of the timing of

birth on completed education, row 2 shows there is no measured difference in mother's wage income between treatment and control groups.[14] The estimated coefficient is statistically indistinguishable from zero, and additionally, we can reject a wage loss of more than .8% at a 95% confidence level. While part of the difference in educational attainment could be from the fact that the treated group appears to be slightly less advantaged, we would expect this to also bias the wage coefficients. The fact that, even despite this potential downward bias, the coefficients are precisely estimated zeros, suggests that, for this population, receiving a high school degree does not have any positive effect on wage income. Row 3 shows that despite the lack of difference in mother's wage income, total family income is lower for treated mothers, and this difference is large and statistically significant for black and Hispanic mothers. Total family income combines the income of all family members, in particular the spouse, which implies either lower spousal earnings for the treated group, or reflects the differences in the probability of marriage observed in Table 5.

The combination of lower family income levels, but a higher number of children means that they are less able to meet their family's needs. Table 7 shows that the treated teenagers are significantly more likely to fall below 100% and 200% of the poverty line. On average, they are 1 percentage point more likely to fall below 100% of the federal poverty line (FPL), and 1.5 percentage points more likely to fall below 200% of the FPL.

It is hard to explain why the birth of a child during high school, rather than one just after high school graduation would result in a lower likelihood of marriage and more children, since this is just a small change in the timing of first birth. However, Lang and Weinstein

[14]The dependent variable in these regressions is log(wage), so women with zero earnings are excluded from the regression. When wage in dollars is used instead, and women with zero earnings are included, the story remains the same. The coefficients are small and statistically indistinguishable from zero.

(2015) find that teenagers who are unmarried at the time of conception get married younger, on average, or not at all. If teenagers who give birth during high school are less likely to get married right away than those who give birth just after high school, it might push more of them into the "not at all" category.

Since the overall family well being appears to be affected by this timing of childbirth, we expect that we might see effects on the children themselves. Children with fewer household resources tend to do worse in school (see, for example: Ananat et al., 2011). We do not have a lot of outcomes available for the children, but we can look at their progression through school. We do this by comparing the children in state s, of age a, in survey year y to the median education level of other similar kids. The result is a indicator variable for whether the child is above or below the median grade for age. This measure has been used in other research (Oreopoulos et al., 2006). Table 8 shows that the timing of birth around the expected graduation date has no estimated effect on the educational progression of this next generation of children.

5.3 Differences Over Time and Over the Lifecycle

High school graduation rates have been increasing over the last several decades. The women who were surveyed in the 1980 Census had their children between 1963 and 1978, while the most recent sample, from the ACS, includes mothers that were surveyed in 2014, so could have had their children as recently as 2012. These young women faced very different environments and opportunities. We include fixed effects for the age of the women and the survey year to capture this heterogeneity in the main results, but in Table 9 through Table 11 we explore some of that heterogeneity. We find that both the high school graduation and years of education effects are strongest in the 1980 Census. For example, for those surveyed in the 1980 census, the treatment group is 9.6 percentage points less likely to have earned a HS degree, and that number falls to 2.6 percentage points for those surveyed in the

2000 census. The magnitude of the coefficient for high school graduation falls even further between the 2000 Census and the ACS, while years of education remains stable over the more recent surveys. It is important to note, here, that Figure 4 included only the the 2000 Census and ACS samples, where the education effects are the smallest. If we were able to produce the same graphs for the 1980 census, where the effect is so large, it's possible we would have observed a discontinuous change in high school graduation. The labor market and family structure outcomes look fairly similar across survey years.

The high school graduation rate here includes recipients of the General Education Development certification. There has been some discussion in the literature (Jaeger and Page, 1996) on whether a GED is as good as a High School diploma. The Decennial data does not distinguish the two paths to completing the high school credential, but the ACS separates them. Table 9 shows that the treated mothers were more likely to earn a GED, rather than a high school diploma, as expected. This implies that if we were able to separate GED recipients from high school graduates in the main results, the effect of the disruption from having a baby prior to high school graduation would be higher than the estimated 7 percent.

Figure 5 shows the coefficients estimated separately by the age of the mother at survey date. These figures are meant to show the effect of teen fertility over the life cycle, though it is important to keep in mind that they could also, in some part, reflect differences in the penalty for giving birth during high school over time. A 35 year old mother, for example, was 35 in either 1980, 2000 or 2005-2014, while a 25 year old mother was 25 in those same years. That means that those who are 35 as of survey date were born, on average, 10 years earlier than those who were 25. Nevertheless, the graphs provide insight on potential lifecycle patterns. Most notably, we can see that the negative effect on high school graduation declines over time. This could reflect the fact that there is likely some "catch-up" in the

form of GED receipt.[15] In addition, we see a small but positive effect on log wage in the early to mid twenties, which declines and then levels out at zero around age 29. This would be consistent with the small differences in the age of the child at each fixed mothers' age and the fact that the ease of working increases as children enter preschool and/or elementary school. The family structure variables do not appear to have any strong patterns.

5.4 Robustness Checks

Section 4 argues that the treatment and control groups are very similar to each other before becoming mothers, particularly after controlling for seasonality with the older cohort. However, even though we have chosen a fairly tight band around the end of high school–approximately six months on either side–the data allow us to test an even tighter band. This should help to alleviate concerns that the control group is not a good comparison. Column (2) of Table 12 show the results of running the same regressions as the main specification, but for the new treatment and control groups. The treatment group only includes those mothers who gave birth during March-June of their senior years, and the control group only includes mothers who gave birth during July-September following their senior years. For the older cohort, mothers who have birth during the 2nd quarter are classified as "treated" and those who gave birth during the 3rd quarter are classified as "control." The coefficients in this column are sometimes smaller, especially for the educational outcomes, as would be expected from Figure 4, but maintain the same patterns as the main results.

We also check the importance of the decision to use women who gave birth between ages 23-25, rather than another age group. We chose ages 23-25 in an attempt to still use relatively young mothers, but ones who are far less likely to experience an interruption in school (including college) due to their pregnancy. Column (3) shows the results using a

[15]Due to sample size restrictions, we are unable to show the results of this separately.

sample of women who became mothers between ages 20-22. This group is much closer in age to the teen mothers. The coefficients in these columns look very similar to the main results. We also used a logistic regression, instead of linear probability model, for the outcome variables that are categorical variables. The coefficients remained consistent in sign and significance[16].

Finally, we check the pattern of the estimates around the change in abortion availability with the Roe v. Wade decision in 1973. Figure 6 shows the lifecycle graphs from section 5.3 with only the observations from the 1980 Decennial Census. This cohort overlapped with the Roe v. Wade decision, which is indicated on the graphs with the vertical line. Women who were younger when they were surveyed in 1980 turned 18 after the Roe v. Wade decision, the women who were older in 1980 turned 18 before the decision. If selective abortion were driving our results, we would expect to see strong effects to the left of the vertical line and no effects to the right. Although the coefficients bounce around, our results do not appear to be driven strongly by the availability of abortions.

6 Conclusions

Lowering teenage pregnancy rates is one of the top priorities for public health officials in the United States. Its correlation with poor economic outcomes for both the teenage mothers and her children makes it an easy target. However, given the disadvantaged backgrounds of teenage mothers when compared to women who delayed childbearing, it is difficult to establish causality. The most convincing previous literature finds that the true causal effect is much lower than correlations would suggest, even when those correlations control for observable measures of family background.

This paper finds that while having a child during high school significantly effects some

[16]Results available on request.

later life outcomes of young mothers. We find that having a child during the last six months of high school causes a seven percent decrease in the probability of obtaining a high school degree, when compared to women who had a child just after the end of high school. It also decreases the probability of marriage and increases the average number of children. However, the disruption to education does not have any measurable effect on earnings. This suggests that the signaling value of a high school degree is not important for this group of young mothers, and that simply helping a teenage mother finish high school will not help improve her earnings potential. Since we find that the timing of births matters for family structure, the families of women who have a child during the last six months of high school are significantly more likely to be living in poverty.

This builds on a larger national discussion on poverty and inequality. How do we improve the outcomes of individuals from poor socioeconomic backgrounds? How do we reduce the cycle of poverty? Unfortunately, this paper does not answer those questions and calls into question two popular policy prescriptions: reducing teenage pregnancy and increasing high school completion. Even though teen mothers with disrupted high school education were less likely to graduate from high school, their later life earnings are consistent with those that had no disruption to their high school progression.

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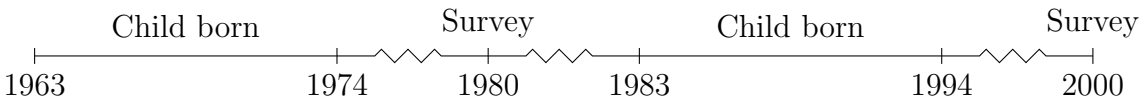
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Figures

Figure 1: Timeline



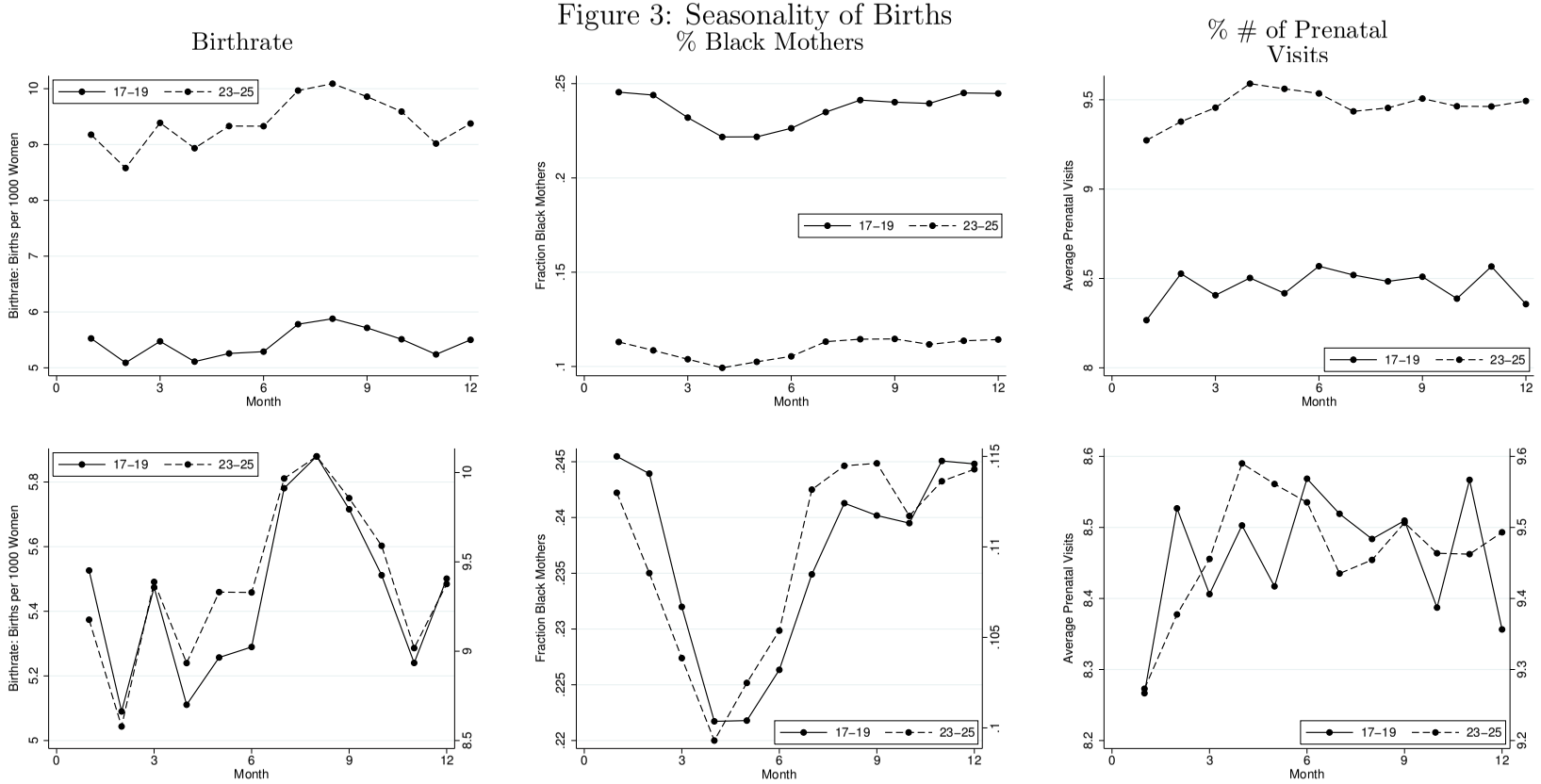
Notes: The timeline shows the relationship between the dates the children are born relative to the survey dates for the mothers in the treated group, for the Decennial Census observations. The young women had their children when they were around age 18 and we observe them when they are 24–35.

Figure 2: Assignment to Treatment and Control Groups

Mom's Quarter of Birth	Child's Quarter of Birth: Treat 1	Treat 2	Control 3	Control 4
1	17: Junior	17: Junior	17: Summer J/S	17: Senior
	18: Senior	18: Senior	18: Summer S/C	18: College1
	19: College1	19: College1	19: Summer C1/C2	19: College2
2	17: Senior	17: Junior	17: Summer J/S	17: Senior
	18: College1	18: Senior	18: Summer S/C1	18: College1
	19: College2	19: College1	19: Summer C1/C2	19: College2
3	17: Senior	17: Senior	17: Summer J/S	17: Senior
	18: College1	18: College1	18: Summer S/C1	18: College1
	19: College2	19: College2	19: Summer C1/C2	19: College2
4	17: Junior	17: Junior	17: Summer J/S	17: Junior
	18: Senior	18: Senior	18: Summer S/C1	18: Senior
	19: College1	19: College1	19: Summer C1/C2	19: College1

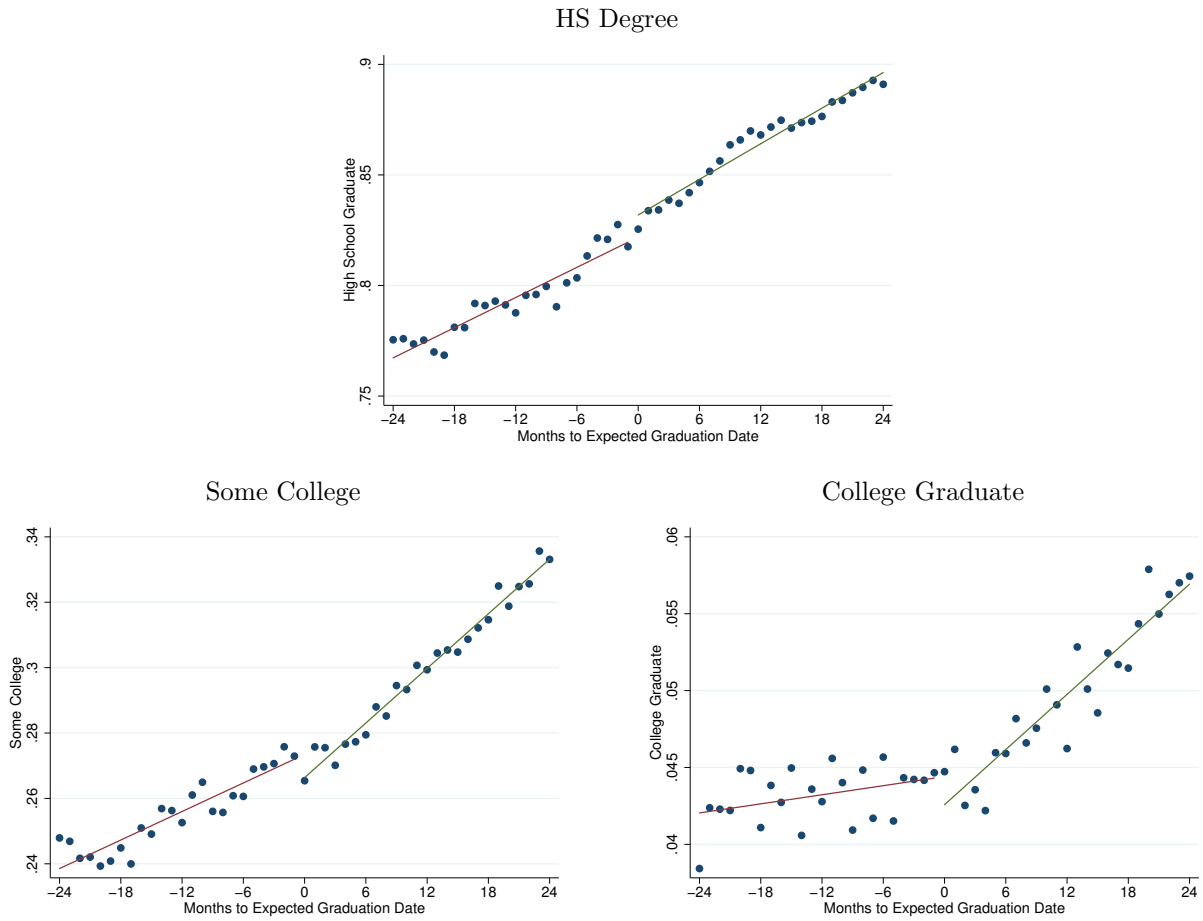
Notes: The shaded rows in the first two columns represent the treated group, while the shaded rows in the 3rd and 4th columns represent the control group. Each cell displays the school year at ages 17, 18 and 19 for someone who has made normal progress in school. "Junior" is the second to last year of high school. "Senior" is the last year of high school. "College1" denotes the first year of college, and "College2" denotes the second year of college. "J/S" is the summer between Junior and Senior year of high school, "S/C1" is the summer between Senior year and the first year of college, and "C1/C2" is the summer between the first two years of college.

Figure 3: Seasonality of Births



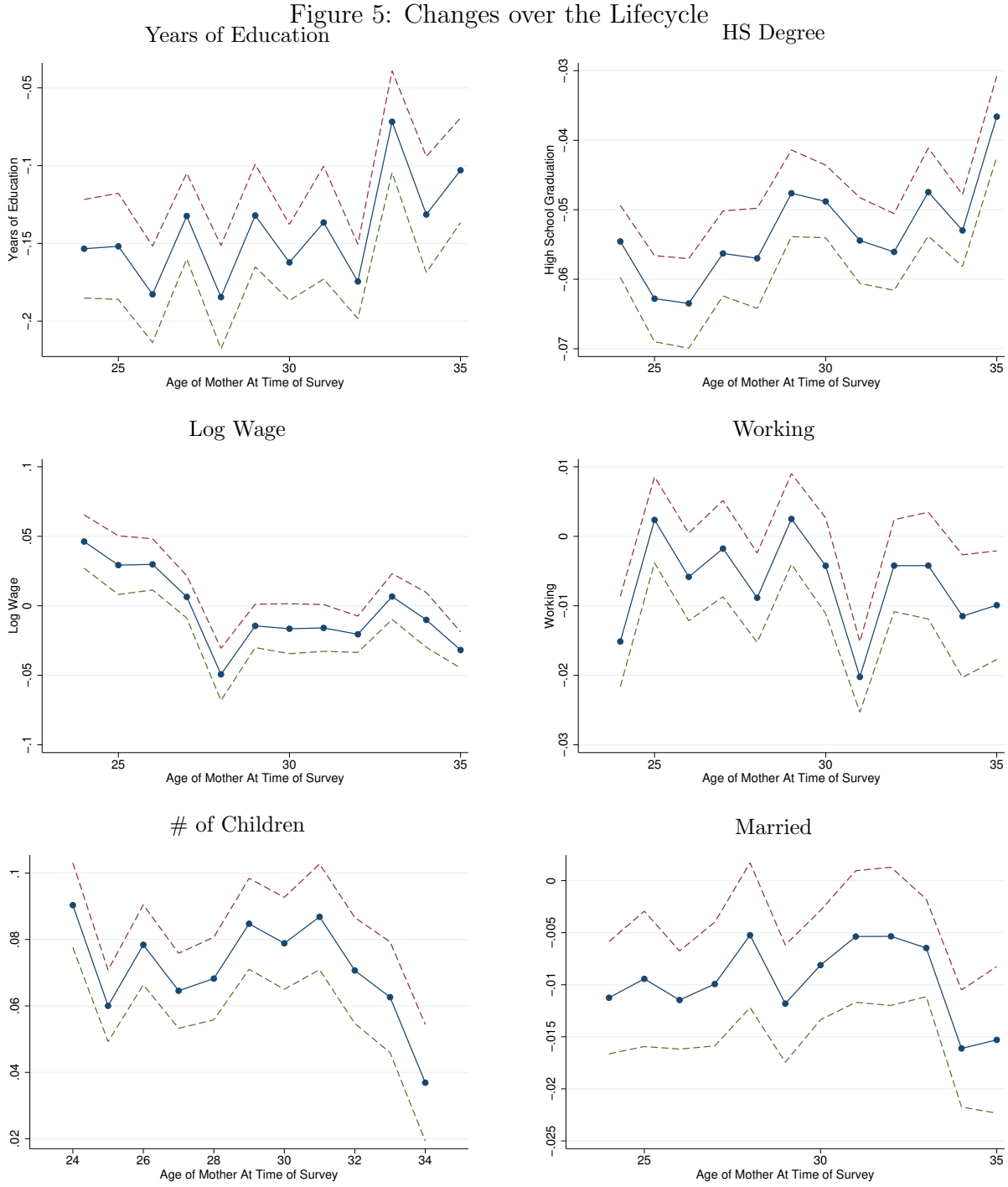
Notes: Figures compiled using the 1970-2012 Natality Data from the National Vital Statistics System of the National Center for Health Statistics. The solid line shows the average, by month of (child's) birth, for 17-19 year old mothers. The dashed line shows the average for 23-25 year old mothers. Figures in the first row show both age groups on the same scale, while figures in the second row use different scales to allow for a better comparison of the seasonal patterns (left axis for 17-19 year olds and right axis for 23-25 year olds).

Figure 4: Birth and Graduation Timing



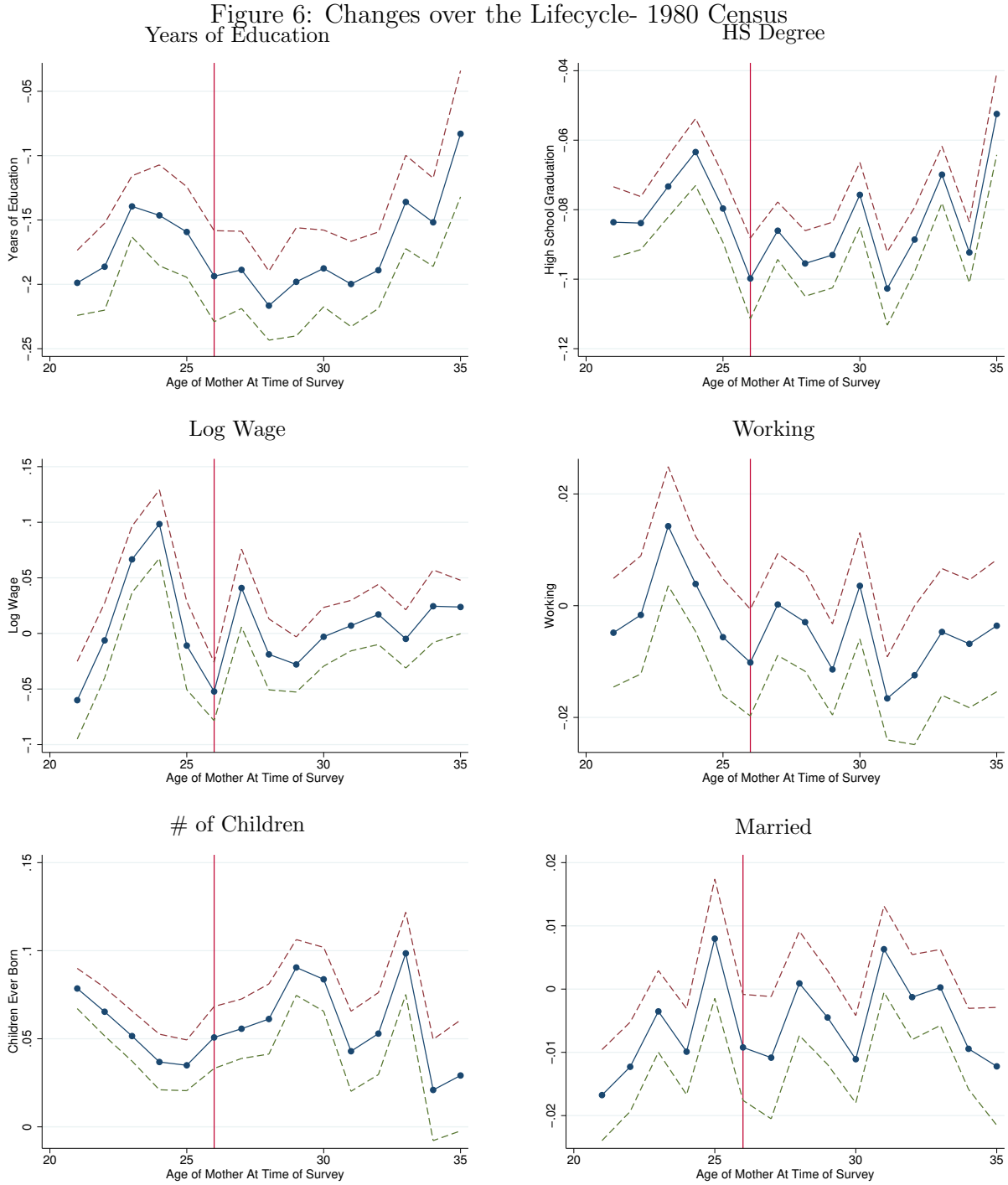
Notes: Each data point is the average value of the variable for women who had their first children t months from their expected graduation date, where t is given on the X-axis. The sample only includes women who were at least 23 years old when surveyed and were part of the 2000 Census and ACS samples.

Figure 5: Changes over the Lifecycle



Notes: Each data point represents the coefficient β_1 from a separate estimation of the following equation: $\text{Outcome}_{is} = \alpha + \beta_1 \text{Treat}_{is} \times \text{Teen}_{is} + \beta_2 \text{Treat}_{is} + \beta_3 \text{Teen}_{is} + \Phi_s + \Phi_r + \varepsilon_{is}$. The regression is estimated separately for each year of mothers' age. The dotted lines give the 95% confidence interval. It is important to note that these graphs also show differences between cohorts. For example, mothers who are aged 35 at survey were (mostly) born in 1945 (1980-35), 1965 (2000-35) and 1970-1979 (ACS year - 35).

Figure 6: Changes over the Lifecycle- 1980 Census



Notes: Each data point represents the coefficient β_1 from a separate estimation of the following equation: $\text{Outcome}_{is} = \alpha + \beta_1 \text{Treat}_{is} \times \text{Teen}_{is} + \beta_2 \text{Treat}_{is} + \beta_3 \text{Teen}_{is} + \Phi_s + \Phi_r + \varepsilon_{is}$. The regression is estimated separately for each year of mothers' age. The dotted lines give the 95% confidence interval. These graphs show only the cohort surveyed in the 1980 Decennial Census. The vertical line separates the women that turned 18 before and after the Roe v. Wade supreme court decision.

Tables

Table 1: Summary Statistics, Full Sample

	Teen		Older	
	Control	Treat	Control	Treat
Age	27.76***	27.64	29.76***	29.74
Black	0.159***	0.175	0.0737***	0.0710
Education (Years)	11.82***	11.67	13.40	13.40
HS Graduate	0.757***	0.704	0.936**	0.938
Wage	10336.4	10346.3	12895.2	12957.4
Total Family Income	39928.1***	39265.8	55159.2**	55379.0
Working	0.547**	0.543	0.565	0.566
Married	0.652***	0.634	0.825***	0.827
Total Children	2.372***	2.431	1.864***	1.869
Log Wage	9.200	9.203	9.442	9.442
Log Total Income	10.28***	10.25	10.68**	10.68
100% Poverty	0.229***	0.244	0.0923	0.0915
200% Poverty	0.516***	0.536	0.267**	0.265
N	200,700	251,100	596,300	622,400

Notes: Data compiled from the 1980 Census, the 2000 Census and the 2005-2014 ACS samples. The first two columns show the averages for the treatment and control groups of a cohort of teenage mothers. The last two columns show the averages for a cohort of women who had their first child between ages 23-25. The stars represent statistically significant differences between the treatment and control groups, within cohort.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 2: NSFG Variables

Variable	Description	Coding
Sexual Activity/Education		
FirstInt	Age at First Intercourse	Age in Months
SexEdNo	Sexual Education: Saying No	1=yes, 0=no
SexEdBC	Sexual Education: Using Birth Control	1=yes, 0=no
VirPledge	Ever took a virginity pledge	1=yes, 0=no
Family Variables		
MomTeenMom	Mother was a Teen Mom	1=yes, 0=no
Intact	Lived in "intact" family: with both parents	1=yes, 0=no
MomHS	Mom has high school degree	1=yes, 0=no
DadHS	Dad has high school degree	1=yes, 0=no
MomWork	Whether mom worked for pay most of the time while she was between ages 5-15	1=yes, 0=no
MomChild	Mom's number of children	Number of Children
Other		
RelServ	Ever attended religious services	1=yes, 0=no
RelServOft	Attended religious services often: >=once a week	1=yes, 0=no
Want	Feelings about timing of pregnancy	1 if too overdue or right time, 0 otherwise
Black	Race	1=black, 0=white

Notes: This table describes the coding for variables from the National Survey of Family Growth.

Table 3: NSFG - Difference-in-Differences

	FirstInt	SexEdNo	SexEdBC	VirPledge	MomTeenMom	Intact	MomHS
Treat × Teen	**-0.146**	**0.0767**	**-0.0537**	**-0.0460**	**0.0187**	**-0.0593**	**0.0025**
	(0.267)	(0.142)	(0.153)	(0.117)	(0.0556)	(0.0522)	(0.0514)
Treat	-0.0010	-	-	-	-0.0454	0.0378	0.0134
	(0.185)	-	-	-	(0.0385)	(0.0362)	(0.0356)
Teen	-2.613***	-	-	-	0.114***	-0.155***	-0.0448
	(0.195)	-	-	-	(0.0407)	(0.0381)	(0.0375)
Mean	17.136	0.684	0.636	0.110	0.396	0.635	0.712
N	1356	40	40	40	1280	1356	1356

	DadHS	MomWork	MomChild	RelServ	RelServOft	Want	Black
Treat × Teen	**-0.0222**	**0.0294**	**-0.112**	**0.0298**	**0.0345**	**-0.0430**	**-0.0003**
	(0.0493)	(0.0522)	(0.181)	(0.162)	(0.202)	(0.0487)	(0.0492)
Treat	0.0317	-0.0582	0.0388	-0.0571	0.0286	0.0263	-0.0283
	(0.0342)	(0.0361)	(0.125)	(0.150)	(0.187)	(0.0338)	(0.0342)
Teen	-0.0431	0.0199	0.169	-0.0463	-0.0985	-0.388***	0.138***
	(0.0359)	(0.0380)	(0.132)	(0.118)	(0.147)	(0.0355)	(0.0358)
Mean	0.729	0.671	3.638	0.829	0.535	0.430	0.165
N	1356	1345	1355	200	200	1356	1255

Notes: The variable listed at the top of each column is the outcome variable used when estimating the following equation: $\text{Outcome}_i = \alpha + \beta_1 \text{Treat}_i \times \text{Teen}_i + \beta_2 \text{Treat}_i + \beta_3 \text{Teen}_i + \varepsilon_i$

The coefficient highlighted in bold is β_1, which gives the difference in differences for the group of teenagers who gave birth right before the end of high school relative to those who gave birth shortly after. Data come from the National Survey of Family Growth, and the outcome variables are defined in Table 2.

* $p < .1$, ** $p < .05$, *** $p < .01$

Table 4: Effect of Teen Fertility on Education

	All	White	Black	Hispanic
High School Graduate	-0.0536***	-0.0631***	-0.0404***	-0.0362***
	(0.00236)	(0.00328)	(0.00335)	(0.00344)
	.734	.744	.787	.629
Some College	-0.00919***	-0.00723***	-0.0191***	-0.0129**
	(0.00148)	(0.00169)	(0.00496)	(0.00458)
	.194	.18	.264	.168
College Graduate	0.00122	0.00242*	0.000721	-0.00698***
	(0.000913)	(0.00107)	(0.00242)	(0.00197)
	.0274	.0261	.0341	.0218
Years of Education	-0.136***	-0.130***	-0.126***	-0.199***
	(0.00824)	(0.00977)	(0.0216)	(0.0226)
	11.75	11.85	12.17	10.83
Obs	1480300	1108800	142200	167000

Notes: Each cell contains the coefficient β_1 from a separate estimation of the following equation: $\text{Outcome}_{isa} = \alpha + \beta_1 \text{Treat}_{isa} \times \text{Teen}_{isa} + \beta_2 \text{Treat}_{isa} + \beta_3 \text{Teen}_{isa} + \Phi_s + \Phi_a + \Phi_r + \varepsilon_{isa}$ The variable listed in the first column is the outcome variable. Mean values of each outcome variable are listed below the coefficient and standard errors. Standard errors are shown in parentheses and are clustered by state. * p<0.05, ** p<0.01, *** p<0.001

Table 5: Effect of Teen Fertility on Family Structure

	All	White	Black	Hispanic
Married	-0.0105***	-0.00705***	-0.0174**	-0.0118***
	(0.00170)	(0.00196)	(0.00542)	(0.00238)
	.644	.734	.325	.625
Number of Children	0.0624***	0.0449***	0.0788***	0.0914***
	(0.00383)	(0.00427)	(0.0101)	(0.00972)
	2.398	2.302	2.517	2.617
Age Gap	0.0949***	0.128***	0.00273	0.0270
	(0.0101)	(0.0121)	(0.0283)	(0.0206)
	3.361	3.362	3.334	3.388
Obs	1670700	1233000	163500	201000

Notes: Each cell contains the coefficient β_1 from a separate estimation of the following equation: $\text{Outcome}_{isarc} = \alpha + \beta_1 \text{Treat}_{isarc} \times \text{Teen}_{isarc} + \beta_2 \text{Treat}_{isarc} + \beta_3 \text{Teen}_{isarc} + \Phi_s + \Phi_a + \Phi_r + \Phi_c + \varepsilon_{isarc}$. The variable listed in the first column is the outcome variable. Mean values of each outcome variable for the treated group are listed below the coefficient and standard errors. Standard errors are shown in parentheses and are clustered by state. * p<0.05, ** p<0.01, *** p<0.001

Table 6: Effect of Teen Fertility on Labor Market Outcomes

	All	White	Black	Hispanic
Working	-0.00676***	-0.00479*	-0.00481	-0.00747
	(0.00178)	(0.00217)	(0.00506)	(0.00411)
	.545	.548	.588	.489
Log Wage	-0.000526	0.00173	-0.00468	0.00178
	(0.00400)	(0.00497)	(0.0102)	(0.0134)
	9.2	9.16	9.31	9.26
Log Total Family Income	-0.0158***	-0.00622	-0.0492***	-0.0212**
	(0.00284)	(0.00325)	(0.0108)	(0.00779)
	10.27	10.4	9.872	10.15
Obs	1651700	1224000	159400	196300

Notes: Each cell contains the coefficient β_1 from a separate estimation of the following equation: $\text{Outcome}_{isarc} = \alpha + \beta_1 \text{Treat}_{isarc} \times \text{Teen}_{isarc} + \beta_2 \text{Treat}_{isarc} + \beta_3 \text{Teen}_{isarc} + \Phi_s + \Phi_a + \Phi_r + \Phi_c + \varepsilon_{isarc}$. The variable listed in the first column is the outcome variable. Mean values of each outcome variable for the treated group are listed below the coefficient and standard errors. Standard errors are shown in parentheses and are clustered by state. * p<0.05, ** p<0.01, *** p<0.001

Table 7: Effect of Teen Fertility on Poverty

	All	White	Black	Hispanic
Below 100% of FPL	0.00989***	0.00508**	0.0203***	0.0130***
	(0.00119)	(0.00162)	(0.00337)	(0.00283)
	.236	.173	.397	.316
Below 200% of FPL	0.0151***	0.0125***	0.0253***	0.0144**
	(0.00151)	(0.00196)	(0.00487)	(0.00469)
	.5246	.4533	.6877	.6374
Obs	1668400	1232500	163300	199900

Notes: Each cell contains the coefficient β_1 from a separate estimation of the following equation: $\text{Outcome}_{isarc} = \alpha + \beta_1 \text{Treat}_{isarc} \times \text{Teen}_{isarc} + \beta_2 \text{Treat}_{isarc} + \beta_3 \text{Teen}_{isarc} + \Phi_s + \Phi_a + \Phi_r + \Phi_c + \varepsilon_{isarc}$. The variable listed in the first column is the outcome variable. Mean values of each outcome variable for the treated group are listed below the coefficient and standard errors. Standard errors are shown in parentheses and are clustered by state. * p<0.05, ** p<0.01, *** p<0.001

Table 8: Effect of Teen Fertility on Grade for Age of Children

	All	White	Black	Hispanic
Below Median Grade For Age	0.00264	-0.00213	0.00132	0.00265
	(0.00322)	(0.00417)	(0.00612)	(0.00332)
Mean	.159	.154	.171	.166

Notes: Each cell contains the coefficient β_1 from a separate estimation of the following equation: $\text{Outcome}_{isarc} = \alpha + \beta_1 \text{Treat}_{isarc} \times \text{Teen}_{isarc} + \beta_2 \text{Treat}_{isarc} + \beta_3 \text{Teen}_{isarc} + \Phi_s + \Phi_a + \Phi_r + \Phi_c + \varepsilon_{isarc}$. The variable listed in the first column is the outcome variable. Mean values of each outcome variable for the treated group are listed below the coefficient and standard errors. Standard errors are shown in parentheses and are clustered by state. * p<0.05, ** p<0.01, *** p<0.001

Table 9: Effect of Teen Fertility on Education, By Survey

	All	1980 Census	2000 Census	ACS
HS Graduate	-0.0528***	-0.0953***	-0.0261***	-0.0139***
	(0.00239)	(0.00388)	(0.00196)	(0.00227)
	.734	.632	.78	.86
GED				0.00475**
				(0.00152)
				.0523
Some College	-0.00826***	-0.0145***	-0.0115**	-0.00292
	(0.00157)	(0.00158)	(0.00381)	(0.00226)
	.194	.111	.299	.249
College Graduate	0.00141	0.0122***	-0.00775***	-0.00745***
	(0.000905)	(0.00139)	(0.00215)	(0.00175)
	.0274	.0172	.0371	.0362
Years of Education	-0.130***	-0.166***	-0.0995***	-0.105***
	(0.00781)	(0.0108)	(0.0114)	(0.0242)
	11.75	11.42	11.91	12.38
Obs	197100	1646500	729500	398900

Notes: Each cell contains the coefficient β_1 from a separate estimation of the following equation: $\text{Outcome}_{isa} = \alpha + \beta_1 \text{Treat}_{isa} \times \text{Teen}_{isa} + \beta_2 \text{Treat}_{isa} + \beta_3 \text{Teen}_{isa} + \Phi_s + \Phi_a + \Phi_r + \varepsilon_{isa}$ The variable listed in the first column is the outcome variable. Mean values of each outcome variable are listed below the coefficient and standard errors. Standard errors are shown in parentheses and are clustered by state. * p<0.05, ** p<0.01, *** p<0.001

Table 10: Effect of Teen Fertility on Labor Market Outcomes, By Survey

	All	1980 Census	2000 Census	ACS
Working	-0.00544**	-0.00935***	-0.00268	-0.00473
	(0.00180)	(0.00237)	(0.00332)	(0.00327)
	.545	.484	.608	.597
Log Wage	0.00416	0.00668	-0.00567	-0.00541
	(0.00410)	(0.00663)	(0.00744)	(0.00753)
	9.2	9.01	9.27	9.41
Log Total Family Income	-0.0154***	-0.00793	-0.0228**	-0.0167**
	(0.00289)	(0.00426)	(0.00690)	(0.00527)
	10.27	10.32	10.14	10.28
Obs	1085900	436700	287000	362200

Notes: Each cell contains the coefficient β_1 from a separate estimation of the following equation: $\text{Outcome}_{isa} = \alpha + \beta_1 \text{Treat}_{isa} \times \text{Teen}_{isa} + \beta_2 \text{Treat}_{isa} + \beta_3 \text{Teen}_{isa} + \Phi_s + \Phi_a + \Phi_r + \varepsilon_{isa}$ The variable listed in the first column is the outcome variable. Mean values of each outcome variable for the treated group are listed below the coefficient and standard errors. Standard errors are shown in parentheses and are clustered by state. * p<0.05, ** p<0.01, *** p<0.001

Table 11: Effect of Teen Fertility on Family Structure, By Survey

	All	1980 Census	2000 Census	ACS
Married	-0.0116***	-0.00827***	-0.0150***	-0.0107**
	(0.00179)	(0.00175)	(0.00360)	(0.00350)
	.644	.754	.595	.509
Number of Children	0.0623***	0.0686***	0.0500***	0.0624***
	(0.00384)	(0.00357)	(0.00962)	(0.00701)
	2.398	2.391	2.331	2.456
Age Gap	0.0968***	0.0858***	0.121***	0.0633**
	(0.0101)	(0.0130)	(0.0206)	(0.0183)
	3.361	3.1	3.45	3.713
Obs	1670700	738000	404600	528000

Notes: Each cell contains the coefficient β_1 from a separate estimation of the following equation: $\text{Outcome}_{isa} = \alpha + \beta_1 \text{Treat}_{isa} \times \text{Teen}_{isa} + \beta_2 \text{Treat}_{isa} + \beta_3 \text{Teen}_{isa} + \Phi_s + \Phi_a + \Phi_r + \varepsilon_{isa}$ The variable listed in the first column is the outcome variable. Mean values of each outcome variable for the treated group are listed below the coefficient and standard errors. Standard errors are shown in parentheses and are clustered by state. * p<0.05, ** p<0.01, *** p<0.001

Table 12: Robustness Checks

	Main	Small Window	Young Control
High School Graduate	-0.0542***	-0.0338***	-0.0456***
	-0.00234	-0.00251	-0.00233
	0.733	0.7267	0.7335
Years of Education	-0.137***	-0.0695***	-0.118***
	-0.00817	-0.0103	-0.00757
	11.75	11.7	11.8
College Grad	0.00122	0.00517***	-0.000817
	(0.000913)	(0.00124)	(0.000631)
	.0274	.0275	.0274
Working	-0.00676***	-0.00662**	-0.00236
	-0.00178	-0.00203	-0.00184
	0.545	0.547	0.545
Log Wage	-0.000526	0.00241	0.00294
	-0.004	-0.00662	-0.00341
	9.2	9.21	9.2
Log Total Family Income	-0.0158***	-0.0121**	-0.0120***
	-0.00284	-0.00411	-0.00324
	10.27	10.26	10.27
Married	-0.0105***	-0.00748***	-0.00755***
	-0.0017	-0.00208	-0.00138
	0.644	0.638	0.644
Number of Children	0.0624***	0.0254***	0.0451***
	-0.00383	-0.00553	-0.00407
	2.398	2.413	2.398
Age Gap	0.0949***	0.0420**	0.109***
	(0.0101)	(0.0126)	(0.0102)
	3.361	3.217	3.361
Below 100% of FPL	0.00989***	0.00604**	0.00696***
	-0.00119	-0.00184	-0.00125
	0.236	0.239	0.236
Below 200% of FPL	0.0151***	0.00886**	0.0122***
	-0.00151	-0.00256	-0.00173
	0.5246	0.5295	0.5246
Obs	1499600	836000	2170700

Notes: Each cell contains the coefficient β_1 from a separate estimation of the following equation: $\text{Outcome}_{isa} = \alpha + \beta_1 \text{Treat}_{isa} \times \text{Teen}_{isa} + \beta_2 \text{Treat}_{isa} + \beta_3 \text{Teen}_{isa} + \Phi_s + \Phi_a + \Phi_r + \varepsilon_{isa}$ The variable listed in the first column is the outcome variable. Mean values of each outcome variable for the treated group are listed below the coefficient and standard errors. Standard errors are shown in parentheses and are clustered by state. * p<0.05, ** p<0.01, *** p<0.001

A Additional Tables

Table A1: Summary Statistics, 1980 Census

	Teen		Older	
	Control	Treat	Control	Treat
Age	27.61***	27.36	29.50***	29.44
Black	0.147***	0.168	0.0604***	0.0575
Education (Years)	11.52***	11.29	13.23***	13.18
HS Graduate	0.674***	0.576	0.929	0.929
College Grad	0.0177	0.0166	0.208***	0.195
Working	0.489***	0.477	0.478	0.477
Married	0.761***	0.744	0.885	0.885
Total Children	2.344***	2.405	1.841***	1.851
Log Wage	9.017	9.010	9.133*	9.122
Wage	8147.6*	8032.9	8840.6	8775.2
Total Family Income	39746.5***	38890.4	51031.5*	50785.9
Log Total Income	10.33***	10.30	10.65	10.64
100% Poverty	0.195***	0.212	0.0778	0.0787
200% Poverty	0.479***	0.505	0.246	0.248
N	94,200	120,900	256,400	266,600

Notes: Data compiled from the 1980 Census. The first two columns show the averages for the treatment and control groups of a cohort teenage mothers. The last two columns show the averages for a cohort of women who had their first child between ages 23-25. The stars represent statistically significant differences between the treatment and control groups, within cohort.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A2: Summary Statistics, 2000 Census

	Teen		Older	
	Control	Treat	Control	Treat
Age	27.54	27.50	30.03	30.04
Black	0.162***	0.173	0.0780*	0.0756
Education	11.94***	11.88	13.27***	13.31
HS Graduate	0.791***	0.766	0.935*	0.937
College Grad	0.0379	0.0361	0.221***	0.227
Wage	11114.6	11222.8	13933.1**	14118.9
Total Family Income	35613.9*	35155.3	51367.5***	52116.7
Working	0.608	0.608	0.633*	0.636
Married	0.603***	0.585	0.816*	0.819
Total Children	2.308***	2.359	1.903*	1.910
Log Wage	9.267	9.272	9.530*	9.540
Log Total Income	10.15**	10.13	10.61***	10.62
100% Poverty	0.269***	0.281	0.103*	0.0998
200% Poverty	0.589***	0.602	0.310***	0.305
N	44,600	54,000	149,700	156,400

Notes: Data compiled from the 2000 Census. The first two columns show the averages for the treatment and control groups of a cohort teenage mothers. The last two columns show the averages for a cohort of women who had their first child between ages 23-25. The stars represent statistically significant differences between the treatment and control groups, within cohort.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A3: Summary Statistics, ACS 2005-2014

	Teen		Older	
	Control	Treat	Control	Treat
Age	28.15	28.17	29.91	29.91
Black	0.177***	0.187	0.0882**	0.0856
Education (Years)	12.41***	12.34	13.87***	13.92
HS Graduate	0.866***	0.853	0.947***	0.951
GED	0.0503***	0.0549	0.0192	0.0192
College Grad	0.0369	0.0354	0.199***	0.206
Wage	13253.2	13230.5	17499.1*	17680.4
Total Family Income	43268.3*	42796.4	63646.6**	64135.2
Working	0.597	0.596	0.627***	0.632
Married	0.515***	0.502	0.751**	0.756
Total Children	2.427***	2.493	1.906	1.908
Log Wage	9.405	9.406	9.724	9.730
Log Total Income	10.29**	10.27	10.77***	10.78
100% Poverty	0.256***	0.266	0.104	0.102
200% Poverty	0.521***	0.535	0.262***	0.257
N	62,000	76,300	190,300	199,500

Notes: Data compiled from the 2005-2014 ACS samples. The first two columns show the averages for the treatment and control groups of a cohort teenage mothers. The last two columns show the averages for a cohort of women who had their first child between ages 23-25. The stars represent statistically significant differences between the treatment and control groups, within cohort.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$